

Safe-Net

Cybersecurity learning with
explainable URL warnings

Artem / Student project / September 2026

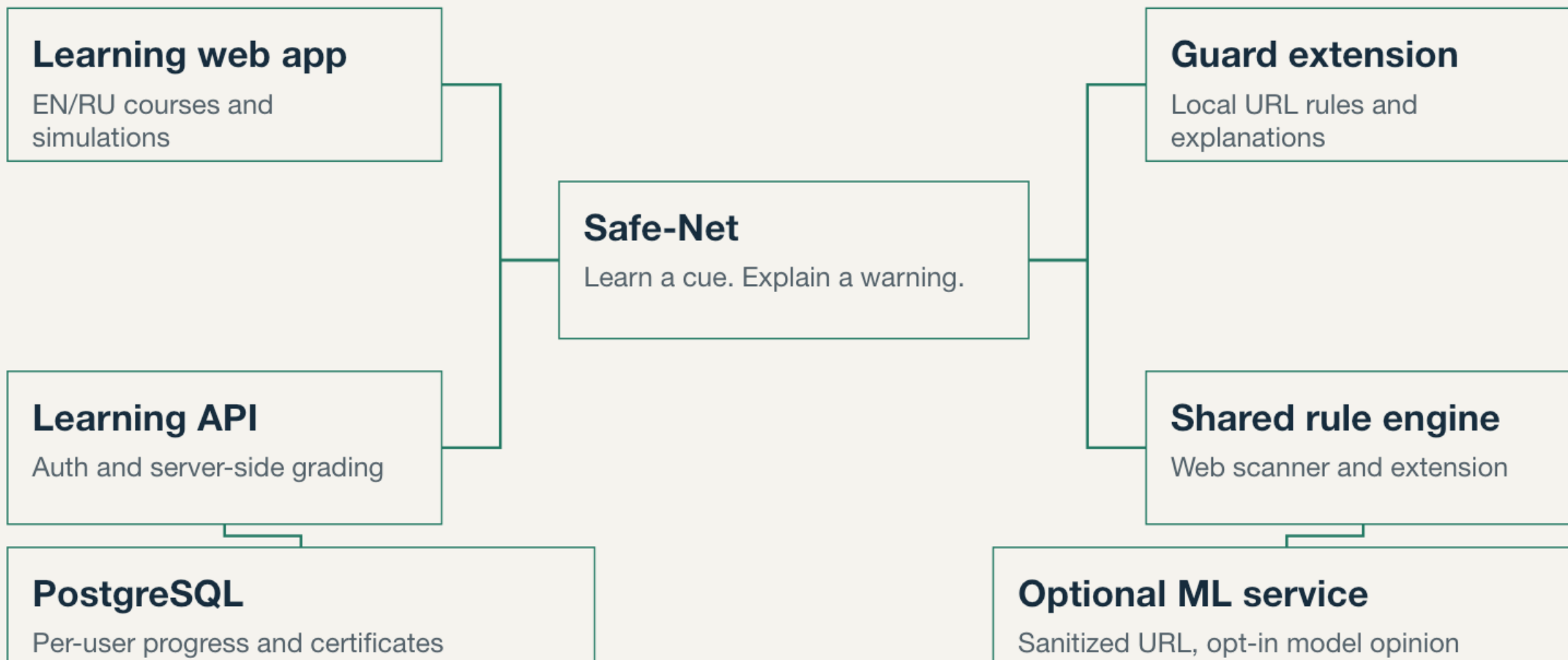
The research question

Can practice with explanations improve recognition of unfamiliar phishing cues?

Proposed comparison

Interactive practice versus an equal-duration text lesson.
Assess new examples immediately and after a delay.

How the system works



The learning experience

21 courses

27 lessons

163 tasks

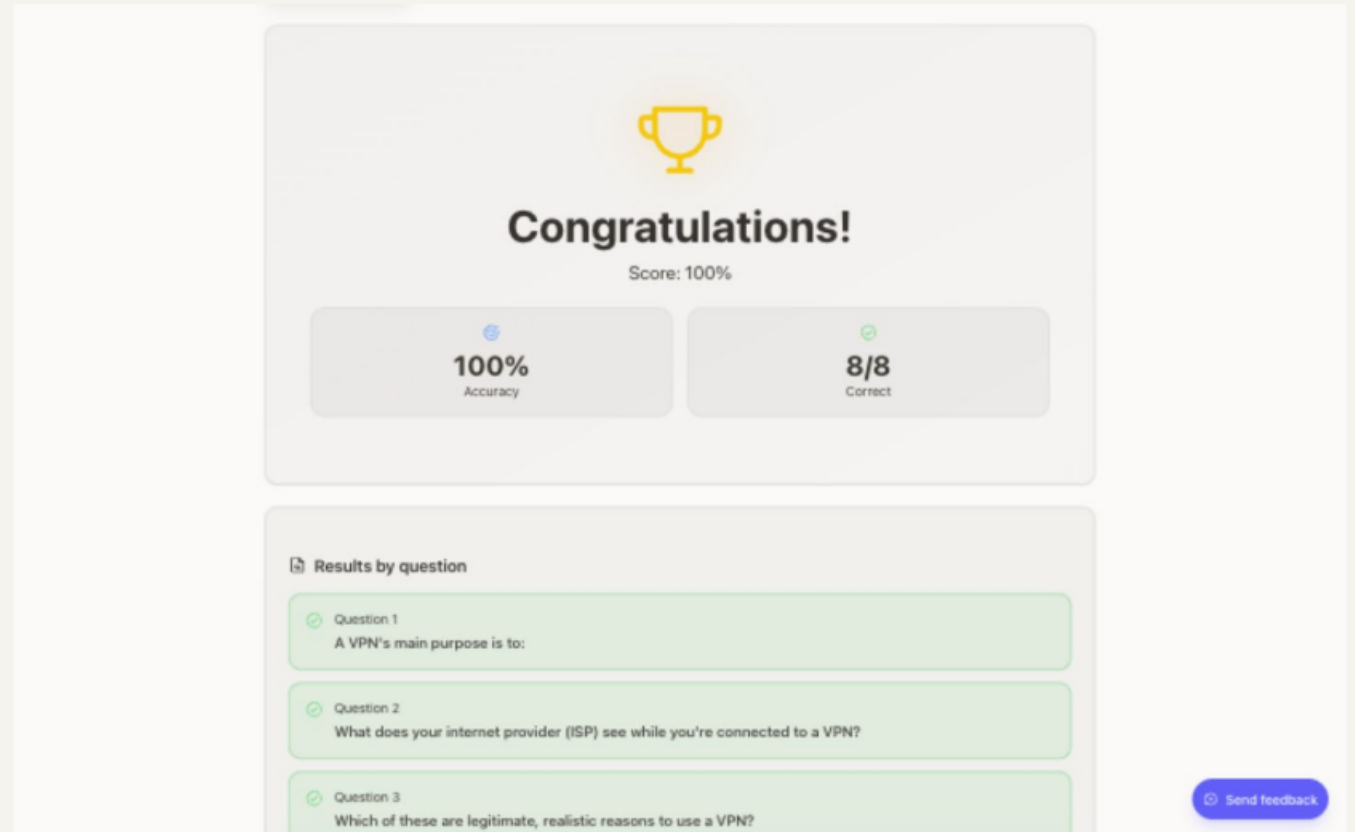
21 final tests

Verified local journey

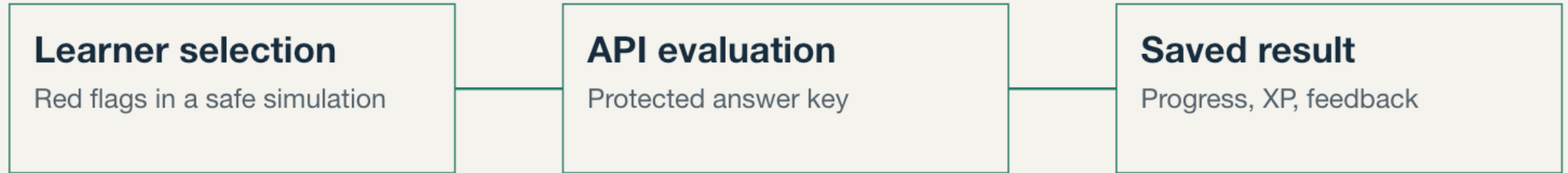
6 tasks completed

8 test answers graded

1 certificate issued



Server-side assessment



A completed task belongs to the learner account

Database constraints prevent repeated XP and duplicate course certificates.
A fresh tab reloads progress from the server.

An explainable warning

micros0ft-alerts.com/login

88 / 100

Danger

Why the rule fires

The domain embeds a brand with a zero in place of “o”. The path also contains a suspicious word.

Risk scores express product policy, not calibrated probabilities.

Privacy and model boundaries

Default: local rules

The browser can score URLs without sending them to the ML service.

Optional: model opinion

Explicit opt-in sends a sanitized URL to the configured endpoint.

What remains in the optional request

Hostnames and paths still reveal information.
The BERT checkpoint comes from a third party.
Independent model evaluation remains open.

Engineering evidence

98

API regression tests

30

URLs in the rule parity corpus

Local builds and package checks passed
HTTP checks covered auth, grading and concurrency
Mailpit received EN/RU account emails

What the evidence does not establish

Research outcomes

- No measured learning gains
- No independent detection accuracy
- No user study yet

Release acceptance

- Chrome installation and behavior
- External inbox delivery and hosted CI
- Accessibility and broader browser coverage

Current scope: a working local prototype

Suitable for technical feedback and study-design review.

Proposed learning-transfer study



Primary outcome

Change in balanced accuracy on delayed, unfamiliar examples

Also report phishing recall, benign false alarms and attrition.
Separate item banks reduce memorization of repeated examples.

A reproducible next contribution

Detector experiment

Build a versioned, independently labelled URL set with source-separated splits.

Error analysis

Compare rules, optional model and blend.
Report false positives by attack family and language.

Deliverable

An evaluation script, documented dataset procedure and a short report that explains failure cases.

Feedback requested

How should I distinguish learning transfer from memorizing familiar phishing cues?

Which assessment design would make a small pilot credible?

Would error analysis or a learning study be the more useful next step?